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United States Patent	12403733
Kind Code	B2
Date of Patent	September 02, 2025
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Vehicle hitching assist apparatus, vehicle hitching assist method, and vehicle hitching assist system

Abstract

A vehicle hitching assist apparatus according to the present invention includes a first mode configured to, when an image pattern of a second hitch portion provided on a trailer is acquired, detect a position of the second hitch portion based on a comparison between the image pattern and an image acquired by an external world perception sensor mounted on a rear portion of a vehicle and output a steering instruction for assisting in hitching between the trailer and the vehicle based on the detected position of the second hitch portion to a steering controller of the vehicle, and a second mode configured to, when the image pattern is not acquired, issue a notification for prompting a driver of the vehicle to manually hitch the trailer and the vehicle.

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Appl. No.:	17/270266
Filed (or PCT Filed):	July 22, 2019
PCT No.:	PCT/JP2019/028610
PCT Pub. No.:	WO2020/039814
PCT Pub. Date:	February 27, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210339588 A1	Nov. 04, 2021

Foreign Application Priority Data

JP 2018-156122 Aug. 23, 2018

Publication Classification

Int. Cl.: B60D1/36 (20060101); B60D1/62 (20060101); B62D15/02 (20060101); G05D1/00 (20240101)

U.S. Cl.:

CPC B60D1/36 (20130101); B60D1/62 (20130101); B62D15/02 (20130101); G05D1/0221 (20130101); G05D1/0231 (20130101)

Field of Classification Search

CPC: B60D (1/36); B60D (1/62); B62D (15/02); B62D (13/06); B62D (15/0285); B60W (10/20); B60W (30/18036); B60W (50/14); B60W (2300/14); G05D (1/0221); G05D (1/0231); G05D (2201/0213); G06V (10/00-2201/136); B60R (1/003)

USPC: 701/28

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a vehicle hitching assist apparatus, a vehicle hitching assist method, and a vehicle hitching assist system that assist in hitching between a trailer and a vehicle.

BACKGROUND ART

(2) PTL1 discloses a vehicle hitching assist apparatus that detects a trailer targeted for hitching, determines a route of a vehicle for hitching this vehicle to the above-described trailer, and controls steering and braking for moving the above-described vehicle along the determined route.

CITATION LIST

Patent Literature

(3) PTL 1: Japanese Patent Application Public Disclosure No. 2016-203972

SUMMARY OF INVENTION

Technical Problem

(4) Then, the assist in the hitching between the vehicle and the trailer requires highly accurate detection of the position of a hitch portion (a hitch coupler) provided on the trailer and a highly accurate movement of the vehicle toward the trailer.

(5) However, although disclosing that the position of the hitch portion provided on the trailer is detected based on an image of a camera, PTL 1 does not disclose a specific method for detecting the position, thereby raising a possibility that the hitching assist fails to sufficiently function due to

an error or a failure in the position detection.

(6) An object of the present invention is to provide a vehicle hitching assist apparatus, a vehicle hitching assist method, and a vehicle hitching assist system capable of highly accurately detecting a position of a hitch portion provided on a trailer, thereby improving hitching accuracy.

Solution to Problem

(7) One aspect of the present invention includes a first mode configured to, when an image pattern of a second hitch portion provided on a trailer is acquired, detect a position of the second hitch portion based on a comparison between the image pattern and an image acquired by an external world perception sensor mounted on a rear portion of a vehicle and output a steering instruction for assisting in hitching between the trailer and the vehicle based on the detected position of the second hitch portion to a steering controller of the vehicle, and a second mode configured to, when the image pattern is not acquired, issue a notification for prompting a driver of the vehicle to manually hitch the trailer and the vehicle.

Advantageous Effects of Invention

(8) According to the one aspect of the present invention, it is possible to highly accurately detect the position of the hitch portion provided on the trailer, thereby improving the hitching accuracy.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 illustrates the configuration of a vehicle including a vehicle hitching assist system.
- (2) FIG. 2 is a side view of a hitch portion between the vehicle and a trailer.
- (3) FIG. 3 is a block diagram of the configuration of a hitching assist control system.
- (4) FIG. 4 is a flowchart illustrating a procedure for a hitching assist.
- (5) FIG. 5 is a flowchart illustrating the procedure for the hitching assist.
- (6) FIG. 6 is a flowchart illustrating the procedure for the hitching assist.
- (7) FIG. 7 is a flowchart illustrating the procedure for the hitching assist.
- (8) FIG. 8 is a flowchart illustrating the procedure for the hitching assist.
- (9) FIG. 9 is a block diagram of a steering control system as the hitching assist.
- (10) FIG. 10 illustrates a coordinate system with the origin thereof placed at a hitch coupler.
- (11) FIG. 11 illustrates a target trajectory in the hitching assist.
- (12) FIG. 12 illustrates a coordinate system with the origin thereof placed at a hitch ball.
- (13) FIG. 13 illustrates a case 1 of coordinate conversion processing.
- (14) FIG. 14 illustrates a case 2 of the coordinate conversion processing.
- (15) FIG. 15 illustrates a case 3 of the coordinate conversion processing.
- (16) FIG. 16 illustrates a case 4 of the coordinate conversion processing.
- (17) FIG. 17 illustrates processing for determining the center of gravity of the vehicle based on the position of the hitch ball.
- (18) FIG. 18 illustrates how to set a midpoint as a control point of a B-spline.
- (19) FIG. 19 illustrates how to set the midpoint as the control point of the B-spline.
- (20) FIG. 20 is a functional block diagram of generation of a steering angle instruction.
- (21) FIG. 21 illustrates processing for calculating a lateral acceleration.
- (22) FIG. 22 illustrates processing for calculating a slip angle and a yaw rate based on a two-wheeled model.
- (23) FIG. 23 illustrates processing for calculating cornering forces F_r and F_r .
- (24) FIG. 24 illustrates processing for updating the position of the vehicle.
- (25) FIG. 25 illustrates processing for updating the target trajectory.
- (26) FIG. 26 exemplarily illustrates a positional relationship that makes the hitching assist highly difficult.

(27) FIG. 27 exemplarily illustrates a positional relationship that makes the hitching assist less difficult.

(28) FIG. 28 is a flowchart illustrating a procedure for the hitching assist.

(29) FIG. 29 is a flowchart illustrating the procedure for the hitching assist.

(30) FIG. 30 is a flowchart illustrating the procedure for the hitching assist.

(31) FIG. 31 is a flowchart illustrating the procedure for the hitching assist.

(32) FIG. 32 is a flowchart illustrating the procedure for the hitching assist.

(33) FIG. 33 is a flowchart illustrating the procedure for the hitching assist.

DESCRIPTION OF EMBODIMENTS

(34) In the following description, embodiments of a vehicle hitching assist apparatus, a vehicle hitching assist method, and a vehicle hitching assist system will be described with reference to the drawings.

(35) FIG. 1 illustrates the configuration of a vehicle 10 including the vehicle hitching assist system according to the present invention.

(36) The vehicle 10 is a four-wheeled vehicle including a pair of front left and right wheels 11, 11 and a pair of rear left and right wheels 12, 12, and is a tow vehicle that tows a trailer 20 (a towed vehicle).

(37) The vehicle 10 and the trailer 20 are hitched via a hitch portion (a hitch) 30.

(38) The vehicle 10 includes an electric power steering apparatus 13, which operates the steering angles of the front wheels 11, 11. The electric power steering apparatus 13 includes a steering actuator such as a motor that generates a steering force, and is a steering apparatus that can autonomously steer the vehicle 10.

(39) A hitching assist control unit 40 is an electronic control apparatus including a microcomputer equipped with a CPU, a ROM, a RAM, and the like, and is a vehicle hitching assist apparatus (a controller) that performs control for assisting in the hitching between the vehicle 10 and the trailer 20.

(40) When a request for the hitching assist is issued from a driver of the vehicle 10, the hitching assist control unit 40 sets a target trajectory, which is a route for moving back the vehicle 10 to a position where the vehicle 10 is hitched to the trailer 20, and sets a target steering angle of the front wheels 11, 11 so as to move back the vehicle 10 along this target trajectory as the hitching assist control.

(41) Then, the hitching assist control unit 40 outputs a signal of the target steering angle (a steering angle instruction) to a steering control unit 14, which is a steering controller that controls the electric power steering apparatus 13 (the steering actuator).

(42) The steering control unit 14 is an electronic control apparatus including a microcomputer equipped with a CPU, a ROM, a RAM, and the like, and controls the steering actuator (the motor) of the electric power steering apparatus 13 in such a manner that the steering angles of the front wheels 11, 11 detected by a front-wheel steering angle sensor 15 become closer to the target steering angle upon receiving the signal of the target steering angle from the hitching assist control unit 40.

(43) In other words, the hitching assist control unit 40 guides the vehicle 10 to the position for the hitch to the trailer 20 by controlling the steering angles of the front wheels 11, 11, thereby providing the assist so as to facilitate the hitching work.

(44) An information display 50, which is a screen display apparatus such as a liquid crystal display, is provided near a driver's seat of the vehicle 10.

(45) Then, the hitching assist control unit 40 has a function of displaying, for example, a notification matter addressed to the driver of the vehicle 10 on the information display 50 as a part of the hitching assist control.

(46) Further, a rear camera 60, which is a camera (an imaging apparatus) that captures an image behind the vehicle 10, is mounted on the rear portion of the vehicle 10 as an external world

perception sensor.

(47) FIG. 2 illustrates the configuration of the hitch portion (the hitch) **30**.

(48) The hitch portion **30** includes a hitch ball **30A** (a first hitch portion) and a hitch coupler **30B** (a second hitch portion). The hitch ball **30A** is disposed at the rear portion of the vehicle **10**. The hitch coupler **30B** is provided at the distal end of an arm **20A** extending forward from the trailer **20**, and is detachably connected to the hitch ball **30A**.

(49) Then, the hitch coupler **30B** holds the hitch ball **30A** so as to cover the upper portion of the hitch ball **30A**, by which the hitch ball **30A** and the hitch coupler **30B** are connected to each other.

(50) The rear camera **60** is mounted on the rear portion of the vehicle **10** with the angle of view, the installation height, the installation angle, and the like thereof adjusted so as to allow the hitch ball **30A** provided on the vehicle **10** to be displayed at a predetermined position on the screen.

(51) FIG. 3 is a block diagram of the configuration of the vehicle hitching assist system.

(52) The hitching assist control unit **40** inputs an image signal from the rear camera **60** and a front-wheel steering angle signal α from the front-wheel steering angle sensor **15**.

(53) Further, the hitching assist control unit **40** inputs a vehicle speed V from a vehicle speed sensor **16**, which detects the running speed (the vehicle speed) of the vehicle **10**, a yaw rate signal γ from a yaw rate sensor **17**, which detects the yaw rate of the vehicle **10**, a gear position signal (a gearshift position signal) GP from a gear position sensor **18**, which detects the gear position (the gearshift position) of the transmission of the vehicle **10**, an output signal from a multi-function switch (a hitching assist operation switch) **19** used for the driver of the vehicle **10** to input an instruction regarding the hitching assist control, and the like.

(54) The multi-function switch **19** is mounted on, for example, a steering wheel **13A** of the vehicle **10**, and is an operation switch such as a press-button switch arbitrarily operated by the driver.

(55) The driver issues an instruction to execute the hitching assist (an instruction to carry out the autonomous steering) and an instruction to initialize the hitching assist (a reset instruction or an instruction to learn an image pattern) by operating the multi-function switch **19**.

(56) For example, the multi-function switch **19** includes a press-button switch labeled "Hitching Assist" and a press-button switch labeled "Initialize".

(57) Then, when the driver presses the "Hitching Assist" button (a driving assist execution switch), a signal indicating the request to execute the hitching assist is output to the hitching assist control unit **40**. When the driver presses the "Initialize" button (a driving assist initialization switch), a signal indicating an instruction to initialize the hitching assist is output to the hitching assist control unit **40**.

(58) The hitching assist control unit **40** includes functions of a subject vehicle behavior calculation portion **40A**, an image processing portion **40B**, a mode setting portion **40C**, a target trajectory generation portion **40D**, and a lateral control portion **40E** in the form of software, as calculation processing functions based on the above-described various kinds of input signals.

(59) The subject vehicle behavior calculation portion **40A** calculates the behavior of the vehicle **10** (a subject vehicle) during the hitching assist based on the vehicle speed signal V , the yaw rate signal γ , a front-wheel steering angle signal α , and the like.

(60) The image processing portion **40B**, for example, extracts the hitch coupler **30B** by processing an image signal from the rear camera **60**.

(61) The mode setting portion **40C** sets a mode of the hitching assist control based on the gear position signal GP , the output signal from the multi-function switch **19** (the instruction regarding the hitching assist from the driver), and the like.

(62) The target trajectory generation portion **40D** generates a target trajectory of the vehicle **10** used when moving back the vehicle **10** to hitch the vehicle **10** to the trailer **20** based on the result of the image processing by the image processing portion **40B** and the mode setting by the mode setting portion **40C**.

(63) The lateral control portion **40E** sets the target steering angle of the front wheels **11**, **11** to move

back the vehicle **10** along the target trajectory based on the result of the image processing by the image processing portion **40B**, the subject vehicle behavior calculated by the subject vehicle behavior calculation portion **40A**, and the target trajectory generated by the target trajectory generation portion **40D**, and outputs a signal of this target steering angle (a steering angle instruction) to the steering control unit **14**.

(64) In the following description, the details of the hitching assist control by the hitching assist control unit **40** will be described with reference to flowcharts illustrated in FIGS. **4** to **8**.

(65) The routine indicated in the flowcharts illustrated in FIGS. **4** to **8** is timer interrupt processing performed every predetermined time (for example, 50 ms).

(66) In step **S101**, the hitching assist control unit **40** performs initialization processing.

(67) At this time, if this is the first time that the hitching assist control unit **40** has been powered on, the hitching assist control unit **40** sets a learning flag FHK for distinguishing whether the image pattern of the hitch coupler **30B** is already learned or not to zero indicating that this image pattern is not learned yet (FHK=0).

(68) Further, if this is the first processing timing after the driving switch of the vehicle **10** (a power switch, an ignition switch, or an engine switch) is switched on (after the vehicle **10** is powered on), the hitching assist control unit **40** resets each of variables P_m and R_m indicating the control state for each gear position to zero indicating the initial state (P_m=0 and R_m=0).

(69) Subsequently, in step **S102**, the hitching assist control unit **40** reads in vehicle information (information indicating the behavior of the subject vehicle) such as the vehicle speed V, the steering angle α , and the gear position signal GP of the vehicle **10**.

(70) Further, in step **S103**, the hitching assist control unit **40** reads in the image captured by the rear camera **60**.

(71) Further, in step **S104**, the hitching assist control unit **40** reads in the instruction signal of the multi-function switch **19**, i.e., the instruction signal indicating the driver's intension regarding the hitching assist (the execution of the hitching assist, the initialization of the hitching assist, or the like).

(72) Then, in step **S105**, the hitching assist control unit **40** determines which position the gear position (the gearshift position) of the vehicle **10** is, a P range (a parking range), an R range (a reverse range), a range other than the P range and the R range (a D range, an N range, and the like).

(73) At this time, the hitching assist control unit **40** proceeds to step **S106**, step **S113**, and step **S131** if the gear position of the vehicle **10** is the P range, is the R range, and is neither the P range nor the R range, respectively.

(74) If the gear position of the vehicle **10** is the P range and the hitching assist control unit **40** proceeds to step **S106**, the hitching assist control unit **40** determines the mode based on the variables P_m and R_m.

(75) The hitching assist control unit **40** proceeds to step **S108** if P_m=0, to step **S107** if P_m=1 and R_m=0, and to step **S117** if P_m=1 and R_m=1.

(76) If P_m=1 and R_m=0 and the hitching assist control unit **40** proceeds to step **S107**, the hitching assist control unit **40** displays the following display content **107** on the information display **50**. Display Content **107**: "Step on the brake and lock the gearshift at the R range." On the other hand, if P_m=0 and the hitching assist control unit **40** proceeds to step **S108**, the hitching assist control unit **40** displays the following two display contents **108-1** and **108-2** on the information display **50**. Display Content **108-1**: "To request the hitching assist, press the "Hitching Assist" button." Display Content **108-2**: "To change the trailer, press the "Initialize" button."

(77) After the display processing in step **S108**, the hitching assist control unit **40** proceeds to step **S109** and determines the operation state of the multi-function switch **19**.

(78) Then, the hitching assist control unit **40** proceeds to step **S110**, step **S112**, and step **S131** if the "Initialize" button is pressed, if the "Hitching Assist" button is pressed, and if the multi-function switch **19** is in a non-operated state without the "Initialize" button and the "Hitching Assist" button

pressed, respectively.

(79) If the “Initialize” button is pressed, i.e., if the instruction for the processing for initializing the driving assist control is issued, in step **S110**, the hitching assist control unit **40** sets the learning flag FHK for distinguishing whether the image pattern of the hitch coupler **30B** is already learned or not to zero indicating that this image pattern is not learned yet (FHK=0).

(80) Subsequently, the hitching assist control unit **40** proceeds to step **S111** and displays the following display content **111** on the information display **50**. Display Content **111**: “The learning state has been initialized.”

(81) Further, the hitching assist control unit **40** proceeds to step **S112**, and sets the variable P_m to 1.

(82) If detecting that the “Hitching Assist” button is pressed in step **S109**, the hitching assist control unit **40** proceeds to step **S112** while skipping steps **S110** and **S111**, and sets the variable P_m in the above-described manner.

(83) After setting the variable P_m in step **S112**, the hitching assist control unit **40** proceeds to step **S131** and performs various kinds of calculation processing prepared for the next occasion of execution of this routine, such as updating a previous value for use in a digital filter calculation, and then ends the present routine.

(84) On the other hand, if detecting that the gear position of the vehicle **10** is the R range in the above-described step, step **S105**, the hitching assist control unit **40** proceeds to step **S113**.

(85) In step **S113**, the hitching assist control unit **40** determines whether the image pattern (the shape) of the hitch coupler **30B** is already learned based on the learning flag FHK.

(86) The hitching assist control unit **40** determines that the image pattern of the hitch coupler **30B** is not learned yet if the learning flag FHK is zero, and determines that the image pattern of the hitch coupler **30B** is already learned if the learning flag FHK is 1.

(87) If the learning flag FHK is zero and the image pattern of the hitch coupler **30B** is not learned yet, the hitching assist control unit **40** proceeds to step **S114**.

(88) In step **S114**, the hitching assist control unit **40** displays the following three display contents **114-1** to **114-3** on the information display **50**. Display Content **114-1**: “The hitching assist is unusable because the learning of the hitching is not completed.” Display Content **114-2**: “Move the vehicle to the hitch position by a manual operation.” Display Content **114-3**: “Lock the gearshift at the P range after moving the vehicle to a hitchable position.”

(89) Presenting the above-described display allows the driver of the vehicle **10** to understand that the vehicle **10** is currently under such a situation that the hitching assist is unusable because the image pattern of the hitch coupler **30B** is not learned yet and further understand that the driver should cause the image pattern of the hitch coupler **30B** to be learned by performing the hitching operation (demonstrating it).

(90) The display processing in the above-described step, step **S114** is a function of issuing a notification prompting the driver of the vehicle **10** to manually hitch the vehicle **10** without the image pattern acquired yet (a second mode).

(91) Subsequently, the hitching assist control unit **40** proceeds to step **S115**, and records the image behind the vehicle **10** that is captured by the rear camera **60**. More specifically, the hitching assist control unit **40** records the image of the rear camera **60** in the course of the driver's manual hitching.

(92) At this time, when recording the image of the rear camera **60**, the hitching assist control unit **40** records the behavior information (the yaw rate, the pitch rate, and/or the like) of the vehicle **10** when the image is captured, in association with the image.

(93) The hitching assist control unit **40** does not have to record all of the images captured by the rear camera **60** in step **S115**, and can save the memory capacity for recording the image captured by the rear camera **60** by recording the image, for example, per predetermined time or per predetermined running distance (for example, 1 m).

(94) Further, the hitching assist control unit **40** proceeds to step **S116** and sets the variable **R_m** to 1, and then proceeds to step **S131**.

(95) If the gear position of the vehicle **10** is the P range when the present routine is performed again after the variable **R_m** is set in step **S116**, the hitching assist control unit **40** proceeds from step **S105** to step **S106** and determines that **R_m=1** and **P_m=1** there, and then proceeds to step **S117**.

(96) In step **S117**, the hitching assist control unit **40** determines whether the hitch ball **30A** is hidden from the image of the rear camera **60** that has been acquired in step **S103** in the present execution of the routine.

(97) The rear camera **60** is set in such a manner that the hitch ball **30A** is displayed on the screen of the rear camera **60**. However, when the hitch ball **30A** and the hitch coupler **30B** are connected to each other, the hitch ball **30A** is supposed to be hidden under the hitch coupler **30B** as the hitch coupler **30B** covers the hitch ball **30A**.

(98) Therefore, when the hitch ball **30A** is hidden from the image, the hitching assist control unit **40** can estimate that the hitch coupler **30B** is manually connected to the hitch ball **30A** and, further, that the hitch coupler **30B** is displayed in the region where the hitch ball **30A** has been displayed.

(99) In other words, the hitching assist control unit **40** can determine based on the image of the rear camera **60** that the hitch ball **30A** and the hitch coupler **30B** are connected to each other and the image region of the hitch coupler **30B** is identified, and the hitching assist control unit **40** is now ready to learn the image pattern of the hitch coupler **30B**.

(100) Therefore, when the hitch ball **30A** and the hitch coupler **30B** are connected to each other, the driver does not have to make the hitching assist control unit **40** aware of it by, for example, operating a button, thereby being less burdened by the operation load.

(101) If the hitch ball **30A** is displayed in the image of the rear camera **60**, i.e., the hitch ball **30A** and the hitch coupler **30B** are separated from each other, the hitching assist control unit **40** proceeds to step **S131** because not reaching a learnable state yet.

(102) On the other hand, when the hitch ball **30A** is hidden from the image of the rear camera **60** with the manual connection completed between the hitch ball **30A** and the hitch coupler **30B**, and the hitching assist control unit **40** becomes ready to perform the processing for learning the image pattern of the hitch coupler **30B**, the hitching assist control unit **40** proceeds to step **S118** and starts the learning processing.

(103) As described above, the hitch ball **30A** hidden from the image of the rear camera **60** in the course of the manual hitching indicates that the manual connection is completed between the hitch ball **30A** and the hitch coupler **30B**, and further indicates that the hitch coupler **30B** is displayed in the image of the rear camera **60** so as to cover and hide the hitch ball **30A**.

(104) Therefore, in step **S118**, the hitching assist control unit **40** performs the processing for learning the image pattern, which extracts the image region (the shape) of the hitch coupler **30B** from the image recorded during the manual hitching based on the position of the hitch ball **30A** in the image of the rear camera **60**, and records this image region into the memory as the image pattern of the hitch coupler **30B**.

(105) After that, the hitching assist control unit **40** repeats processing that dates back from the last image toward the first image among the images of the rear camera **60** that have been recorded in the course of the hitching by the driver's operation (in the course of the manual hitching) and extracts a region shaped similarly to the image region of the hitch coupler **30B** extracted in the immediately preceding image from the present image as the image region of the hitch coupler **30B**, thereby learning (recording) the image pattern (the image region) of the hitch coupler **30B** for each of a plurality of conditions corresponding to different distances from the vehicle **10**.

(106) In other words, the hitching assist control unit **40** repeats processing that searches for the hitch coupler **30B** to newly acquire the image pattern thereof from the image that the hitching assist control unit **40** temporally dates back to based on the acquired image pattern of the hitch coupler **30B**, starting from the image when the hitch ball **30A** is hidden, thereby acquiring the image

patterns under the plurality of conditions corresponding to the different distances from the rear camera **60** to the hitch coupler **30B**.

(107) In this manner, when the driver moves back the vehicle **10** by steering it and hitches the vehicle **10** and the trailer **20**, the hitching assist control unit **40** records the image acquired by the rear camera **60** until the hitch is completed, and acquires the image pattern of the hitch coupler **30B** from the recorded image.

(108) Then, the hitching assist control unit **40** can determine whether to learn the image pattern or cancel the learning based on the behavior information (the yaw rate, the pitch rate, and/or the like) of the vehicle **10** that is recorded together with the image of the rear camera **60**.

(109) More specifically, when a change in the behavior of the vehicle **10** is larger than a predetermined level (the yaw rate and/or the pitch rate is(are) higher than the setting), this leads to a reduction in the image quality of the rear camera **60** such as occurrence of a blur in the image, thereby making it impossible for the hitching assist control unit **40** to accurately learn the image pattern of the hitch coupler **30B**.

(110) Therefore, the hitching assist control unit **40** cancels the learning if the change in the behavior of the vehicle **10** is larger than the predetermined level when recording the image of the rear camera **60**, while learning the image pattern based on the high-quality image captured when the change in the behavior of the vehicle **10** is smaller than the predetermined level, thereby preventing the reduction in the accuracy of the learned data of the image pattern.

(111) The processing for learning the image pattern in the above-described step, step **S118** does not require the driver to select the image region of the hitch coupler **30B** by, for example, operating the touch panel, thereby reducing the operation load on the driver.

(112) Further, this processing sequentially learns the image pattern at a position farther away from the vehicle **10** by repeating the processing that searches for the region of the hitch coupler **30B** based on the extracted region in the immediately preceding image, starting from the image when the hitch ball **30A** is hidden (when the connection is completed), thereby being able to highly accurately and collectively learn the image pattern of the hitch coupler **30B** for each of the conditions corresponding to the different distances from the vehicle **10** based on the hitching manually demonstrated only once.

(113) Subsequently, the hitching assist control unit **40** proceeds to step **S119**, and determines whether or not the image pattern of the hitch coupler **30B** can be learned for each of the conditions corresponding to the different distances from the vehicle **10**, i.e., whether the learning of the image pattern is completed.

(114) Then, if the learning of the image pattern is completed, the hitching assist control unit **40** proceeds to step **S120**, and sets the learning flag FHK to 1 indicating that the image pattern is already learned (FHK=1).

(115) Further, after displaying the following display content **121** on the information display **50** in the next step, step **S121**, the hitching assist control unit **40** proceeds to step **S131**. Display Content **121**: “The learning is completed. The hitching assist is usable from the next time.”

In other words, the hitching assist control unit **40** makes the driver aware that the hitching assist control unit **40** has completed the learning of the image pattern of the hitch coupler **30B** and is now ready to execute the hitching assist based on the detection of the position of the hitch coupler **30B** with use of the image pattern due to the display content **121**.

(116) On the other hand, if the learning of the image pattern is not completed yet, the hitching assist control unit **40** proceeds to step **S122** and displays the following display contents **122-1** and **122-2** on the information display **50**, and then proceeds to step **S131** after that. Display Content **122-1**: “Because the learning has failed, the assist is still unusable next time.” Display Content **122-2**: “Try relearning after changing the sunshine condition or while moving back the vehicle with a milder operation”.

(117) As a reason for the failure in completing the learning, possible examples thereof include the

image captured with reduced sensitivity due to low illuminance, occurrence of a blur in the image due to a sharp behavior of the vehicle **10**, and the like. Therefore, notifying the driver of the above-described display contents can prompt the driver to perform the manual hitching operation under a further suitable condition, thereby guiding the driver to the completion of the learning.

(118) In other words, when failing to acquire the image pattern of the hitch coupler **30B**, the hitching assist control unit **40** issues a notification prompting the driver of the vehicle **10** to change the condition of the manual hitching by displaying the above-described display content **122-2** on the information display **50**.

(119) When proceeding to step **S113** after completing the learning of the image pattern of the hitch coupler **30B**, the hitching assist control unit **40** determines that the image pattern is already learned (already acquired) and performs substitution processing of the variable $R_m=P_m$, and then proceeds to step **S123**.

(120) In step **S123**, the hitching assist control unit **40** determines which value the variable R_m has, 0, 1, or 2.

(121) If the variable R_m is 0 at this time, the hitching assist control unit **40** proceeds to step **S131**.

(122) The variable R_m is set to $R_m=2$ upon the generation of the target trajectory for moving the vehicle **10** toward the position for the hitch to the trailer **20**, and indicates that the target trajectory is not generated when being in a state of $R_m=1$.

(123) Therefore, when $R_m=1$, the hitching assist control unit **40** proceeds to step **S124** and steps subsequent thereto and performs processing for generating the target trajectory.

(124) First, in step **S124**, the hitching assist control unit **40** extracts the trailer **20** from inside the image of the rear camera **60**. More specifically, the hitching assist control unit **40** extracts a combination of two opposing vertical edges and horizontal edges between these vertical edges as the trailer **20**.

(125) Then, when being able to detect one pair of vertical edges and the horizontal edges between them, the hitching assist control unit **40** detects a region most highly correlated to the image pattern of the hitch coupler **30B** from the image region between the one pair of vertical edges as the image region of the hitch coupler **30B** based on pattern matching using the learned image pattern of the hitch coupler **30B**, and calculates the position and the angle of the hitch coupler **30B**.

(126) Subsequently, the hitching assist control unit **40** proceeds to step **S125** and generates the target trajectory for moving back the vehicle **10** (the subject vehicle) to the trailer **20**.

(127) At this time, the hitching assist control unit **40** identifies the position of the subject vehicle and the angle of the subject vehicle in a coordinate system with the origin thereof placed at the position of the hitch coupler **30B**, and generates the target trajectory for moving back the vehicle **10** to position the hitch ball **30A** closer to the hitch coupler **30B**.

(128) Further, in the generation of the target trajectory, the hitching assist control unit **40** defines the hitch ball **30A** as a starting point and the hitch coupler **30B** as an end point and further sets a midpoint based on, for example, the relative relationship between the longitudinal axis of the subject vehicle and the longitudinal axis of the trailer, calculates interpolation points between these control points with use of a B-spline curve, and extracts the interpolation points, thereby generating the target trajectory.

(129) The generation of the target trajectory by the hitching assist control unit **40** will be described in detail below.

(130) After generating the target trajectory, the hitching assist control unit **40** proceeds to step **S126** and sets the variable R_m indicating the R-range state to 2. After that, the hitching assist control unit **40** proceeds to step **S131**.

(131) Due to this setting, when proceeding to step **S123** next time or after that, the hitching assist control unit **40** determines that $R_m=2$ and proceeds to step **S127**.

(132) In step **S127**, the hitching assist control unit **40** calculates a target lateral acceleration G_y for following the target trajectory based on the slip angle β , the vehicle speed V , and the like of the

vehicle **10**.

(133) Subsequently, in step **S128**, the hitching assist control unit **40** calculates a steering instruction based on the target lateral acceleration G_y , and outputs the signal of the target steering angle (the steering angle instruction) to the steering control unit **14**, which controls the electric power steering apparatus **13**.

(134) Further, the hitching assist control unit **40** updates the position of the subject vehicle in step **S129**, and, further updates the position of the hitch ball **30A** in step **S130**.

(135) The processing for calculating the steering angle instruction and updating the position of the subject vehicle and the position of the hitch ball **30A** by the hitching assist control unit **40** will be described in detail below.

(136) In this manner, without the image pattern of the hitch coupler **30B** learned yet, the hitching assist control unit **40** prompts the driver of the vehicle **10** to manually demonstrate the hitching, and learns the image pattern (the shape) of the hitch coupler **30B** based on the image of the rear camera **60** that is acquired when the hitching is manually demonstrated.

(137) Then, once learning the image pattern of the hitch coupler **30B**, the hitching assist control unit **40** becomes ready to execute the hitching assist, and execute the hitching assist when the assist request is issued from the driver.

(138) During the hitching assist, the hitching assist control unit **40** perceives the hitch coupler **30B** from the image of the rear camera **60** based on the learned image pattern of the hitch coupler **30B**, and controls the steering angles of the front wheels **11**, **11** so as to move back the vehicle **10** toward the perceived hitch coupler **30B**.

(139) Therefore, the vehicle **10** is highly accurately moved back as far as the position at which the hitch ball **30A** is connected to the hitch coupler **30B** under the steering control by the hitching assist control unit **40**, and allows the driver to easily hitch the vehicle **10** and the trailer **20** even when there is the driver alone.

(140) In the following description, the processing for generating the target trajectory (step **S125**) and the processing for generating the steering angle instruction (step **S128**), which are performed by the hitching assist control unit **40**, will be described in detail.

(141) Processing in steps **S123** to **S128** illustrated in FIG. **8** is, in summary, a function of detecting the position of the hitch coupler **30B** from the image of the rear camera **60** based on the pre-acquired image pattern of the hitch coupler **30B**, and outputting the steering instruction for assisting in the hitching between the vehicle **10** and the trailer **20**, i.e., the steering instruction for moving the hitch ball **30A** of the vehicle **10** toward the hitch coupler **30B** of the trailer **20** based on the detected position of the hitch coupler **30B** to the steering control unit **14** (the steering controller) (a first mode).

(142) FIG. **9** is a block diagram of a system that controls the steering as the hitching assist.

(143) In the hitching assist system illustrated in FIG. **9**, the rear camera **60** outputs the signal of the captured image to the hitching assist control unit **40**.

(144) The hitching assist control unit **40** includes functions as a hitch coupler detection portion **41**, a target trajectory generation portion **42**, and a steering angle instruction generation portion **43** in the form of software.

(145) The hitch coupler detection portion **41** perceives the hitch coupler **30B** based on a comparison between the pre-learned image pattern of the hitch coupler **30B** and the image of the rear camera **60** to identify the position and the angle of the hitch coupler **30B**, and outputs information about the identified position and angle of the hitch coupler **30B** to the target trajectory generation portion **42**.

(146) The target trajectory generation portion **42** generates the target trajectory of the subject vehicle for connecting the hitch ball **30A** of the subject vehicle to the hitch coupler **30B** of the trailer **20** based on, for example, the information about the position and the angle of the hitch coupler **30B**, and outputs information about the target trajectory to the steering angle instruction

generation portion **43**.

(147) The steering angle instruction generation portion **43** determines the steering angle instruction for causing the subject vehicle to follow the target trajectory, i.e., the steering instruction for assisting in the hitching between the vehicle **10** and the trailer **20**, and outputs the determined steering angle instruction (the steering instruction) to the steering control unit **14** (the steering controller), which controls the electric power steering apparatus **13**.

(148) In the following description, the processing for generating the target trajectory by the target trajectory generation portion **42** will be described in detail.

(149) FIG. **10** illustrates how to establish a coordinate system of the target trajectory.

(150) The target trajectory generation portion **42** sets an orthogonal coordinate system with the origin thereof placed at the position of the hitch coupler **30B** of the trailer **20** and the coordinate axes thereof laid on the longitudinal direction of the trailer **20** (the X axis) and the lateral direction of the trailer **20** (the Y axis), respectively.

(151) In the above-described coordinate system, the angle is expressed in such a manner that an angle toward the positive direction of the Y axis is expressed as a positive angle and an angle toward the negative direction of the Y axis is expressed as a negative angle, assuming that the positive direction of the X axis corresponds to 0 degrees.

(152) Therefore, in the case of the vehicle position illustrated in FIG. **10**, an angle θ of the longitudinal axis of the vehicle **10** is expressed as a negative value ($-90 \text{ degrees} < \theta < 0$).

(153) FIG. **11** illustrates the outline of the processing for generating the target trajectory.

(154) Upon receiving the result of the perception (the position and the angle) of the hitch coupler **30B** from the hitch coupler detection portion **41**, the target trajectory generation portion **42** sets the coordinate system with the origin thereof placed at the position of the hitch coupler **30B** (refer to FIG. **10**), and starts expressing the position and the angle of the subject vehicle in this coordinate system.

(155) Next, the target trajectory generation portion **42** sets a plurality of control points of the B-spline curve for defining the shape of the target trajectory, and calculates interpolation points between the control points according to the B-spline curve.

(156) The target trajectory generation portion **42** sets a position shifted forward from the hitch coupler **30B** by a predetermined distance in the direction of the longitudinal axis of the trailer **20** (the end point-1) and a position shifted backward from the hitch ball **30A** by a predetermined distance in the direction of the longitudinal axis of the vehicle **10** (the starting point+1), and, further, the midpoint, which is the intermediate position between the “end point-1” and the “starting point+1”, in addition to the position of the hitch ball **30A**, which is the starting point, and the position of the hitch coupler **30B**, which is the end point, as the control points.

(157) Then, the target trajectory generation portion **42** extracts the interpolation points at equal-distance intervals, thereby generating the target trajectory formed by the extracted interpolation points.

(158) Now, the input information for the generation of the target trajectory will be described.

(159) The hitch coupler detection portion **41** expresses the position of the hitch coupler **30B** (x, y) in an orthogonal coordinate system with the origin thereof placed at the position of the hitch ball **30A** and the coordinate axes thereof laid on the longitudinal direction of the vehicle **10** (the X axis) and the lateral direction of the vehicle **10** (the Y axis), respectively, as illustrated in FIG. **12**.

(160) Further, the hitch coupler detection portion **41** expresses an angle Φ of the trailer **20** so as to express an angle toward the positive direction of the Y axis as a positive angle and an angle toward the negative direction of the Y axis as a negative angle, assuming that the positive direction of the X axis corresponds to 0 degrees, similarly to the coordinate system of the target trajectory.

(161) Then, the hitch coupler detection portion **41** outputs information about the position of the hitch coupler **30B** (x, y) and the angle Φ of the trailer **20** ($\Phi > 0$ in FIG. **12**) in the coordinate system with the origin thereof placed at the position of the hitch ball **30A** illustrated in FIG. **12** to the target

trajectory generation portion **42**.

(162) Because the coordinate system is established in a different manner from the hitch coupler detection portion **41**, the target trajectory generation portion **42** performs processing for converting the positional information so as to express the position of the hitch coupler **30A** (X, Y) in the coordinate system with the origin thereof placed at the position of the hitch coupler **30B**.

(163) In the above-described conversion processing, the target trajectory generation portion **42** performs different conversion processing according to the positional relationship between the vehicle **10** and the trailer **20**.

(164) FIG. **13** (a case 1) is a case in which there is no intersection point between the longitudinal axis of the trailer **20** and the longitudinal axis of the vehicle **10**, i.e., a case in which the longitudinal axis of the trailer **20** and the longitudinal axis of the vehicle **10** extend in parallel with each other at some interval, and the angle Φ (rad) of the longitudinal axis of the trailer **20** is zero.

(165) In this case, when the position of the hitch coupler **30B** in the coordinate system with the origin thereof placed at the position of the hitch ball **30A** is expressed as (x, y), the position of the hitch ball **30A** (X, Y) in the coordinate system with the origin thereof placed at the position of the hitch coupler **30B** is converted into (X=x, Y=y), as illustrated in FIG. **13**.

(166) FIG. **14** (a case 2) is a case in which there is an intersection point between the longitudinal axis of the trailer **20** and the longitudinal axis of the vehicle **10** beyond (behind) the trailer **20**.

(167) In this case, assuming that x1 represents the distance (m) from the hitch coupler **30B** to the intersection point along the longitudinal axis of the vehicle **10** and X1 represents the distance (m) from the hitch coupler **30B** to the intersection point along the longitudinal axis of the trailer **20**, the following equations are established.

$$y/x1 = \tan \Phi$$

$$y/X1 = \sin \Phi$$

$$(X+X1)/(x+x1) = \cos \Phi$$

Therefore, when the position of the hitch coupler **30B** in the coordinate system with the origin thereof placed at the position of the hitch ball **30A** is (x, y), the position of the hitch ball **30A** (X, Y) is converted into (X=x cos Φ - y sin Φ , Y=x sin Φ + y cos Φ) in the coordinate system with the origin thereof placed at the position of the hitch coupler **30B**.

(168) FIG. **15** (a case 3) is a case in which there is an intersection point between the longitudinal axis of the trailer **20** and the longitudinal axis of the vehicle **10** in front of the vehicle **10**.

(169) In this case, assuming that x1 represents the distance from the hitch ball **30A** to the intersection point along the longitudinal axis of the vehicle **10** and X1 represents the distance from the hitch ball **30A** to the intersection point along the longitudinal axis of the trailer **20**, the following equations are established.

$$y/(x+x1) = \tan \Phi$$

$$X1/x1 = \cos \Phi$$

$$(x+x1)/(X+X1) = \cos \Phi$$

Therefore, when the position of the hitch coupler **30B** in the coordinate system with the origin thereof placed at the position of the hitch ball **30A** is (x, y), the position of the hitch ball **30A** (X, Y) is converted into (X=x cos Φ + y sin Φ , Y=-x sin Φ + y cos Φ) in the coordinate system with the origin thereof placed at the position of the hitch coupler **30B**.

(170) FIG. **16** (a case 4) is a case in which there is an intersection point between the longitudinal axis of the trailer **20** and the longitudinal axis of the vehicle **10** within the longitudinal position x.

(171) In this case, assuming that x2 represents the distance from the hitch ball **30A** to the intersection point along the longitudinal axis of the vehicle **10**, x1 represents the distance from the intersection point to the hitch coupler **30B** along the longitudinal axis of the vehicle **10**, X1 represents the distance (m) from the hitch coupler **30B** to the intersection point along the longitudinal axis of the trailer **20**, and X2 represents the distance (m) from the intersection point to the hitch ball **30A** along the longitudinal axis of the trailer **20**, the following equations are

established.

$$y/x1 = \tan \Phi$$

$$x2 = x - x1$$

$$y/X1 = \sin \Phi$$

$$Y/X2 = \tan \Phi$$

Therefore, when the position of the hitch coupler **30B** in the coordinate system with the origin thereof placed at the position of the hitch ball **30A** is (x, y), the position of the hitch ball **30A** (X, Y) is converted into ($X = x \cos \Phi + y \sin \Phi$, $Y = x \sin \Phi - y \cos \Phi$) in the coordinate system with the origin thereof placed at the position of the hitch coupler **30B**.

(172) The target trajectory generation portion **42** identifies the position of the hitch ball **30A** (X, Y) in the coordinate system with the origin thereof placed at the position of the hitch coupler **30B** in this manner.

(173) Subsequently, the target trajectory generation portion **42** identifies the position of the center of gravity (Xg, Yg) of the vehicle **10** in the coordinate system with the origin thereof placed at the position of the hitch coupler **30B** (refer to FIG. 17).

(174) At this time, because the vehicle **10** has the yaw angle θ (rad) of $\theta = \Phi$, the target trajectory generation portion **42** calculates the position of the center of gravity (Xg, Yg) according to the following equations, assuming that Lb represents the distance (m) from the center of gravity of the vehicle to the hitch ball **30A**.

$$Xg = X + Lb \cdot \text{Math.cos } \theta$$

$$Yg = Y + Lb \cdot \text{Math.sin } \theta$$

(175) Next, the setting of the control points of the B-spline curve will be described.

(176) The target trajectory generation portion **42** sets the following first to fifth control points (the starting point to the end point) as the control points of the B-spline curve. the first control point (the starting point): the position of the center of gravity of the vehicle **10** the second control point (the starting point+1): a position shifted backward from the position of the center of gravity of the vehicle **10** by a predetermined distance (for example, 2 m) along the direction of the longitudinal axis of the vehicle **10** the third control point (the midpoint): the intersection point or the central point between the longitudinal axis of the vehicle **10** and the longitudinal axis of the trailer **20** the fourth control point (the end point-1): a position shifted forward from the hitch coupler **30B** by a predetermined distance (for example, 3.5 m) along the direction of the longitudinal axis of the trailer **20** the fifth control point (the end point): the position of the hitch coupler **30B**

(177) At this time, the target trajectory generation portion **42** selects any one of the intersection point and the central point between the longitudinal axis of the vehicle **10** and the longitudinal axis of the trailer **20** as the third control point (the midpoint) in the following manner.

(178) The central point refers to an intersection point between diagonal lines of a rectangle having a diagonal line that is a line connecting the second control point (the starting point+1), which is set at the position shifted backward from the position of the center of gravity of the vehicle **10** by the predetermined distance in the direction of the longitudinal axis of the vehicle **10**, and the fourth control point (the end point-1), which is set at the position shifted forward from the hitch coupler **30B** by the predetermined distance in the direction of the longitudinal axis of the trailer **20**, as illustrated in FIGS. 18 and 19.

(179) The target trajectory generation portion **42** selects the intersection point as the third control point (the midpoint) if the intersection point between the longitudinal axis of the vehicle **10** and the longitudinal axis of the trailer **20** is contained in the above-described rectangle as illustrated in FIG. 18.

(180) On the other hand, the target trajectory generation portion **42** selects the central point, which is the intersection point between the diagonal lines of the rectangle, as the third control point (the midpoint) if the intersection point between the longitudinal axis of the vehicle **10** and the longitudinal axis of the trailer **20** is not contained in the above-described rectangle as illustrated in

FIG. 19.

(181) The B-spline curve does not necessarily pass through the control points, but has to pass through the starting point and the end point in the generation of the target trajectory by the target trajectory generation portion **42**.

(182) Therefore, the target trajectory generation portion **42** allows the B-spline curve to pass through the starting point and the end point by setting the control point next to the starting point (the second control point or the starting point+1) at the position shifted backward from the position of the center of gravity, which is the starting point, by the predetermined distance along the direction of the longitudinal axis of the vehicle **10**, and also setting the control point immediately before the end point (the fourth control point or the end point-1) at the position shifted forward from the hitch coupler **30B** as the end point by the predetermined distance along the direction of the longitudinal axis of the trailer **20**.

(183) Next, the definition of the B-spline curve will be described now.

(184) A B-spline curve of order n including one segment is formed by $n+1$ control points $\{P_0, P_1, \dots, P_n\}$, and a B-spline function of order n including L segments is defined by an equation 1.

(185)
$$P(t) = \sum_{i=0}^{n+L-1} P_i N_i^n(t) \quad [\text{Equation } 1]$$

(186) The range in which the parameter t changes is t_n to t_n+L .

(187) A quadratic basis function $N_{\text{sub}.0.\text{sup}.2}(t)$ of a uniform quadratic B-spline curve is defined by an equation 2.

(188)
$$N_0^2(t) = \begin{cases} \frac{1}{2}t^2 & t \in [0, 1] \\ -t^2 + 3t - \frac{3}{2} & t \in [1, 2] \\ \frac{1}{2}(t-3)^2 & t \in [2, 3] \\ 0 & t \in [0, 3] \end{cases} \quad [\text{Equation } 2]$$

(189) Another quadratic basis function $N_{\text{sub}.i.\text{sup}.2}(t)$ is acquired by translating $N_{\text{sub}.0.\text{sup}.2}(t)$ by i in the positive direction of the t axis.

(190) Now, the interval between the interpolation points according to the B-spline curve depends on the interval between the control points and a step size dt of the parameter t , and the interval between the interpolation points cannot be known before the calculation.

(191) Therefore, the step size dt of the parameter t is preset so as to allow the interpolation points to be calculated at sufficiently short intervals according to the requirement of the target trajectory even when the interval between the control points is different.

(192) In the present embodiment, the target trajectory is generated by extracting the interpolation points according to the B-spline curve at intervals of 0.1 m in terms of the longitudinal position. This necessitates the step size dt that keeps the maximum interval between the interpolation points shorter than 0.1 m even when the interval between the control points is different, and the step size dt is determined to be 0.01 as a value satisfying this requirement.

(193) After calculating the interpolation points according to the B-spline curve, the target trajectory generation portion **42** generates the target trajectory as exemplarily illustrated in FIG. **11** by extracting the interpolation points at intervals of 0.1 m in terms of the longitudinal position.

(194) However, because there are no interpolation points located at equal-distance intervals, the target trajectory generation portion **42** acquires the interpolation points at equal-distance intervals by conducting linear interpolation between interpolation points preceding and subsequent to the position at the equal-distance interval.

(195) Next, the function of the steering angle instruction generation portion **43** illustrated in FIG. **9** will be described in detail.

(196) FIG. **20** is a functional block diagram of the steering angle instruction generation portion **43**.

(197) The steering angle instruction generation portion **43** includes a coordinate conversion portion

43A, a single-point preview LK controller 43B, a proportional FB controller 43C, a steering angle conversion portion 43D, a vehicle model portion 43E, and a vehicle position update portion 43F.

(198) The single-point preview LK controller 43B calculates the lateral acceleration for causing the vehicle 10 to follow the target trajectory.

(199) The single-point preview LK controller 43B calculates the lateral acceleration G_y required when the vehicle 10 moves back and runs by a distance L_p (m) in the longitudinal direction (the front-rear direction) and is further displaced laterally by a distance e_y (m) as illustrated in FIG. 21, hypothetically supposing that the vehicle 10 corners in a steady state with the vehicle speed V (m/s) and the front-wheel tire steering angle δ (rad) kept constant.

(200) Now, a vehicle longitudinal speed V_x (m/s) and a vehicle lateral speed V_y (m/s) when the vehicle 10 moves back are acquired as the vehicle longitudinal speed $V_x = V \cdot \cos \beta$ and the vehicle lateral speed $V_y = V \cdot \sin \beta$, respectively, based on the vehicle speed V and the slip angle β (rad).

(201) On the other hand, a preview time T_p (s) is acquired as $T_p = L_p / V_x$ based on the vehicle longitudinal speed V_x and the distance L_p .

(202) Further, the lateral displacement e_y is acquired as the lateral displacement $e_y = V_y \cdot T_p + \frac{1}{2} \cdot G_y \cdot T_p^2$ based on the vehicle lateral speed V_y , the preview time T_p , and the lateral acceleration G_y .

(203) This means that the lateral acceleration G_y is acquired as the lateral acceleration $G_y = \{2(e_y - V_y \cdot T_p)\} / T_p^2$.

(204) The single-point preview LK controller 43B calculates the lateral acceleration G_y by assuming that the preview distance is constant and setting the lateral displacement e_y based on the target trajectory.

(205) Then, the single-point preview LK controller 43B causes the vehicle 10 to follow the target trajectory by setting the lateral acceleration G_y for each control cycle.

(206) An increase in the preview distance L_p leads to a reduction in the lateral displacement at the position of the hitch coupler 30B but an excessive increase in the preview distance L_p causes an overshoot, and therefore the preview distance L_p is adapted based on the convergence to the position of the hitch coupler 30B. In the present embodiment, the preview distance L_p is set to 4.0 m based on the result of a simulation.

(207) The steering angle conversion portion 43D calculates a steering-wheel steering angle δ_h based on the lateral acceleration G_y with use of a calculation equation of steady-state cornering in a two-wheeled model.

(208) The yaw rate γ (rad/s) is calculated according to an equation 3 based on a wheelbase L_w (m) of the vehicle 10, a stability factor A (s.sup.2/m.sup.2) of the vehicle 10, the vehicle speed V (m/s), and the front-wheel tire steering angle δ (rad).

(209)
$$\gamma = \frac{1}{1 + AV^2} \frac{V}{L_w} \quad [\text{Equation } 3]$$

(210) Further, because of the lateral acceleration $G_y = V \cdot \gamma$ and $\gamma = G_y / V$, the front-wheel tire steering angle δ is calculated according to an equation 4.

(211)
$$\delta = \frac{(1 + AV^2)L_w}{V^2} G_y \quad [\text{Equation } 4]$$

(212) Then, assuming that G_r represents an overall gear ratio, the steering-wheel steering angle δ_h is acquired as $\delta_h = G_r \cdot \delta$ based on the front-wheel tire steering angle δ and the overall gear ratio G_r .

(213) The vehicle model portion 43E calculates the slip angle β and the yaw rate γ according to the two-wheeled model.

(214) In a state that the vehicle 10 moves forward illustrated in FIG. 22, “the mass×the lateral acceleration=the lateral external force” and “the inertial moment x the angular speed=the moment around the center of gravity due to the external force” are established.

(215) Therefore, an equation of motion when the vehicle **10** moves forward is expressed as an equation 5, assuming that m represents the inertial mass.

$$m(\dot{v} + uy) = F_{sub.f} + f_{sub.r}$$

$$I\dot{\gamma} = l_{sub.f}F_{sub.f} - l_{sub.r}F_{sub.r} \quad [\text{Equation 5}]$$

(216) Now, focusing on a motion in a range where the slip angle β of the vehicle body is small when the vehicle **10** moves back, the following relationships are established because the direction of the lateral speed when the vehicle **10** moves back has an opposite sign from when the vehicle **10** moves forward.

$$u \approx V$$

$$v = -V \sin \beta \approx -V\beta$$

(217) Further, because $+F_f$ and $+F_r$ are generated with respect to the front and rear-wheel slip angles $+\beta_f$ and $+\beta_r$ when the vehicle **10** moves back, a motion equation when the vehicle **10** moves back is expressed as an equation 6.

$$mV(\dot{\beta}) - \gamma = F_{sub.f} - F_{sub.r}$$

$$I\dot{\gamma} = l_{sub.f}F_{sub.f} - l_{sub.r}F_{sub.r} \quad [\text{Equation 6}]$$

(218) The motion equation when the vehicle **10** moves back expressed as the equation 6 means that the slip angle β and the yaw rate γ can be calculated if F_f and F_r are identified.

(219) In the following description, processing for calculating F_f and F_r by the vehicle model portion **43E** will be described with reference to FIG. **23**.

(220) In FIG. **23**, the front-wheel tire steering angle δ = the steering-wheel steering angle δ_h / G_r is established, and, further, the vehicle longitudinal speed $V_x = V \cos \beta$ and the vehicle lateral speed $V_y = V \sin \beta$ are acquired based on the vehicle speed V and the slip angle β .

(221) Further, the front-wheel slip angle β_f and the rear-wheel slip angle β_r are calculated according to the following equations based on the vehicle longitudinal speed V_x , the vehicle lateral speed V_y , a distance L_f between the center of gravity and the front wheel, a distance L_r between the center of gravity and the rear wheel, the yaw rate γ , and the front-wheel tire angle δ .

$$\beta_f = \tan^{-1}\{(V_y + L_f \gamma) / V_x\} - \delta$$

$$\beta_r = \tan^{-1}\{(V_y - L_r \gamma) / V_x\}$$

(222) Further, lateral forces S_f and S_r on the front and rear wheels are acquired as $S_f = +2 \cdot K_f \cdot \beta_f$ and $S_r = +2 \cdot K_r \cdot \beta_r$, respectively, based on the slip angles β_f and β_r and front and rear-wheel cornering power K_f and K_r ($N \cdot \text{rad}$).

(223) The $+$ signs are added in the equations for calculating the lateral forces S_f and S_r because $+F_f$ is generated with respect to $+B_f$.

(224) Then, the cornering forces F_f and F_r are acquired as $F_f = S_f \cos \delta$ and $F_r = S_r$, respectively, based on the lateral forces S_f and S_r calculated in the above-described manner and the front-wheel tire angle δ .

(225) At this time, the vehicle model portion **43E** calculates the slip angle β and the yaw rate γ by substituting the calculated cornering forces F_f and F_r into the equation 6.

(226) The vehicle position update portion **43F** inputs information about the slip angle β and the yaw rate γ calculated by the vehicle model portion **43E**.

(227) Then, the vehicle position update portion **43F** calculates the center of gravity (X_{g_t} , Y_{g_t}) and the yaw angle θ_t (refer to FIG. **24**) of the vehicle **10** at time t in the coordinate system with the origin thereof placed at the position of the hitch coupler **30B** according to an equation 7.

$$\theta_{sub.t} = \theta_{sub.0} + \int_0^t \gamma_{sub.t} dt$$

$$X_{sub.g_t} = X_{sub.g_0} + \int_0^t \cos(\beta_{sub.t} + \theta_{sub.t}) V_{sub.0} dt, Y_{sub.g_t} = Y_{sub.g_0} + \int_0^t \sin(\beta_{sub.t} + \theta_{sub.t}) V_{sub.0} dt \quad [\text{Equation 7}]$$

(228) Further, the vehicle position update portion **43F** calculates the position of the hitch ball **30A** (X_{b_t} , Y_{b_t}) at time t in the coordinate system with the origin thereof placed at the position of the hitch coupler **30B**.

(229) At this time, assuming that L_b (m) represents the distance between the center of gravity of the

vehicle and the hitch ball **30A**, the position of the hitch ball **30A** (Xb_t , Yb_t) is calculated according to the following equations.

$$Xb_t = Xg_t - Lb \cdot \cos \theta t$$

$$Yb_t = Yg_t - Lb \cdot \sin \theta t$$

(230) The coordinate conversion portion **43A** inputs information about the position of the center of gravity of the vehicle **10** (Xg_t , Yg_t), the position of the hitch ball **30A** (Xb_t , Yb_t), and the yaw angle θt at time t that are calculated by the vehicle position update portion **43F**.

(231) Then, the coordinate conversion portion **43A** updates the target trajectory in the vehicle coordinate system so as to cancel off the change amount of the position and the change amount of the posture in one control cycle by translating each point on the target trajectory by movement amounts ΔX and ΔY of the vehicle **10** between the control cycles and further rotationally moving each point on the target trajectory by a posture change amount $\Delta \theta$ of the vehicle **10** between the control cycles for each control cycle, as illustrated in FIG. 25.

(232) In other words, the coordinate conversion portion **43A** calculates the change amount of the posture and the movement amount of the vehicle **10** from the starting point and converts the coordinates of the initial target trajectory, thereby setting the target trajectory in the vehicle coordinate system.

(233) It is also possible to recreate the target trajectory by newly setting control points and interpolating points according to the B-spline curve for each control cycle at this time, but a rotational movement at a small angle frequently occurs and calculation errors are accumulated in the hitching assist, so that the present embodiment converts the coordinates of the initial target trajectory and sets the target trajectory in the vehicle coordinate system for each control cycle.

(234) Then, the coordinate conversion portion **43A** outputs information about the updated target trajectory to the single-point preview LK controller **43B** and the proportional FB controller **43C**.

(235) The proportional FB controller **43C** performs feedback control on the steering angle based on a proportional operation, which calculates a lateral difference e (m) between the vehicle position and the target trajectory, and calculates a proportional FB steering angle δP_FB based on this lateral difference e (m) and a proportional gain constant Gp (rad/m).

(236) Then, the proportional FB controller **43C** outputs information about the calculated proportional FB steering angle δP_FB to the steering control unit **14**.

(237) At this time, the proportional FB controller **43C** sets a reference position for calculating the lateral difference e to the position of the hitch ball **30A**.

(238) Setting the reference position for calculating the lateral difference e to the position of the hitch ball **30A** can reduce a change in the trajectory around the hitch coupler **30B** compared to when setting the reference position for calculating the lateral difference e to the position of the center of gravity of the vehicle **10**.

(239) Further, the proportional FB controller **43C** can limit the section of the proportional control on the steering angle (a section in which the feedback control is performed) to around the hitch coupler **30B**. More specifically, the proportional FB controller **43C** stops the proportional control (the feedback control) until the distance from the vehicle **10** to the hitch coupler **30B** falls below a set value, and performs the proportional control (the feedback control) after the distance from the vehicle **10** to the hitch coupler **30B** falls below the set value.

(240) However, an acquired verification result indicates that an overshoot occurs around the hitch coupler **30B** to similar degrees between when the proportional control is performed in the entire range of the target trajectory and when the proportional control is performed only around the hitch coupler **30B**. Therefore, in the present embodiment, the proportional FB controller **43C** performs the proportional control in the entire range of the target trajectory.

(241) Further, the proportional gain constant Gp is a value adapted when the hitching assist control unit **40** is designed and developed so as to improve the convergence to the position of the hitch coupler **30B**, and is set to, for example, approximately 3.0 (rad/m).

(242) Then, a success rate of the hitching assist depends on the positional relationship between the vehicle **10** and the trailer **20** when the hitching assist is started (an initial state).

(243) When the trailer **20** is considerably laterally positionally misaligned from the region behind the vehicle **10** as illustrated in FIG. **26**, such a positional relationship raises the necessity of moving back the vehicle **10** while significantly wiggling it, thereby making the hitching assist difficult. On the other hand, when the trailer **20** is located in the region behind the vehicle **10** as illustrated in FIG. **27**, such a positional relationship requires only a generally linear backward movement of the vehicle **10**, thereby making the hitching assist less difficult.

(244) The driver of the vehicle **10** can be aware of the state that the hitching assist is less difficult with the trailer **20** located in the region behind the vehicle **10** as illustrated in FIG. **27** as a state that the entire trailer **20** targeted for the hitching can be visually confirmed in a rearview mirror **10A**.

(245) Then, the hitching assist control unit **40** can increase the success rate of the hitching assist if executing the hitching assist while employing the establishment of such a positional relationship that the driver can visually confirm the entire trailer **20** targeted for the hitching in the rearview mirror **10A** of the vehicle **10** as a start condition.

(246) Further, if the image pattern of the hitch coupler **30B** is learned with the image quality of the rear camera **60** kept in good condition, this can lead to an increase in the accuracy of the perception of the hitch coupler **30B** by the hitching assist control unit **40**, thereby leading to an increase in the success rate of the hitching assist.

(247) For this reason, the hitching assist control unit **40** can increase the success rate of the hitching assist if learning the image pattern of the hitch coupler **30B** when the image quality of the rear camera **60** is kept in good condition.

(248) In the following description, a second embodiment will be described with reference to flowcharts illustrated in FIGS. **28** to **33**. The above-described condition for starting the hitching assist and condition for learning the image pattern are added in the second embodiment.

(249) The hitching assist control unit **40** performs similar processing in steps **S201** to **S207** in the flowcharts illustrated in FIGS. **28** to **33** to the above-described steps, steps **S101** to step **S107** in the flowcharts illustrated in FIGS. **4** to **8**, and therefore the detailed description of the content of the processing in steps **S201** to **S207** will be omitted here.

(250) In step **S208**, the hitching assist control unit **40** controls the display on the information display **50**.

(251) In the above-described step, step **S108**, the hitching assist control unit **40** displays the display content **108-1**: "To request the hitching assist, press the "Hitching Assist" button." as the display control on the information display **50**.

(252) On the other hand, the hitching assist control unit **40** displays the following display content **208-1** on the information display **50** instead of the above-described display content **108-1**. Display Content **208-1**: "To request the hitching assist, press the "Hitching Assist" button after confirming that you can view the trailer in the rearview mirror." In other words, the hitching assist control unit **40** starts the hitching assist upon satisfaction with such a condition that the trailer **20** targeted for the hitching is positioned so as to be viewed in the rearview mirror **10A**, i.e., that the trailer **20** targeted for the hitching is displayed around the center (a predetermined region) of the image of the rear camera **60**.

(253) As a result, the present configuration can reduce the load of the image processing on the rear camera **60** in the hitching assist processing, and, further, can reduce the necessity of moving back the vehicle **10** while considerably steering the vehicle **10**, thereby making the hitching assist less difficult and thus improving the success rate of the hitching assist.

(254) Further, the hitching assist control unit **40** allows the driver to easily understand that the hitching assist is started provided that the trailer **20** is positioned so as to be able to be viewed in the rearview mirror **10A**, and the success rate of the hitching assist is improved by satisfying this start condition, by displaying the above-described display content **208-1**.

(255) Further, the hitching assist control unit **40** can prompt the driver to perform the operation of moving the vehicle **10** to a position at which the trailer **20** can be viewed in the rearview mirror **10A**, i.e., a preparation operation for causing the hitching assist to start before starting the hitching assist by displaying the above-described display content **208-1**.

(256) The hitching assist control unit **40** displays a display content **208-2** on the information display **50** in step **S208**, with the display content **208-2** indicating the same content as the display content **108-2**: "To change the trailer, press the "Initialize" button." in step **S108**.

(257) The hitching assist control unit **40** performs similar processing in the next steps, steps **S209** to **S214** to the above-described steps, steps **S109** to step **S114** in the flowcharts illustrated in FIGS. **4** to **8**, and therefore the detailed description of the content of the processing in steps **S209** to **S214** will be omitted here.

(258) In step **S215**, the hitching assist control unit **40** records the image behind the vehicle **10** that is captured by the rear camera **60** similarly to the above-described step, step **S115**, but, further records the detected values of the yaw rate and the longitudinal G i.e., the behavior of the vehicle **10** synchronized with the recorded image.

(259) The physical amount indicating the behavior of the vehicle **10** that is recorded together with the recorded image in step **S215** is not limited to the yaw rate and the longitudinal G, and can be a physical amount regarding the behavior of the vehicle, such as the steering angle, the vehicle speed, a change in the steering angle, and a change in the vehicle speed.

(260) The hitching assist control unit **40** can make an image correction for reducing the influence of the behavior based on the direction of this vehicle behavior and the images preceding and subsequent thereto at the time of such a behavior scene that the image quality is deteriorated, by storing information about the physical amount indicating the behavior of the vehicle **10**, such as the yaw rate and the longitudinal G, in synchronization with the recorded image.

(261) For example, capturing an image under a condition that the longitudinal G is large leads to extension of the screen in the vertical direction. In this case, if identifying a change in the pitch angle per G in advance, the hitching assist control unit **40** can determine how many degrees the pitch angle is changed based on a change in the G with respect to a sampling interval.

(262) Then, the hitching assist control unit **40** can acquire a corrected image in which the influence of the longitudinal G is reduced by converting the change in the pitch angle into the number of pixels and correcting the image so as to reduce the extension in the vertical direction.

(263) Further, when the image is captured under a condition that the yaw rate is high, the hitching assist control unit **40** can correct extension of the screen in the horizontal direction based on the information about the yaw rate pre-recorded in synchronization with the recorded image.

(264) Regarding the content of processing in the next steps, steps **S216** to **S218**, the hitching assist control unit **40** performs similar processing to the above-described steps, steps **S116** to step **S118** in the flowcharts illustrated in FIGS. **4** to **8**, and therefore the detailed description of the content of the processing in steps **S216** to **S218** will be omitted here.

(265) In step **S219**, the hitching assist control unit **40** determines whether the learning of the image pattern of the hitch coupler **30B** is completed.

(266) In the above-described step, step **S119**, the hitching assist control unit **40** determines that the learning is completed if the image pattern can be learned for each condition with respect to all of the conditions corresponding to the different distances from the vehicle **10**.

(267) On the other hand, in step **S219**, the hitching assist control unit **40** determines whether all of the vehicle behaviors recorded in step **S215** satisfy the following determination conditions in addition to whether the image pattern can be learned for each condition with respect to all of the conditions corresponding to the different distances from the vehicle **10**. $\text{abs}(\text{the yaw rate}) < \text{a first predetermined value}$ $\text{abs}(\text{the longitudinal G}) < \text{a second predetermined value}$

(268) In these conditions, $\text{abs}(A)$ is a function that returns an absolute value of a variable A .

(269) Then, if being able to learn the image pattern for each distance with respect to all of the

distances and all of the recorded vehicle behaviors also satisfy the determination conditions, the hitching assist control unit **40** determines that the learning is completed and proceeds to step S220.

(270) In other words, the hitching assist control unit **40** acquires the image pattern of the hitch coupler **30B** from the image acquired by the rear camera **60** when a change in the behavior of the vehicle **10** is smaller than a predetermined level.

(271) On the other hand, if the above-described learning completion condition is not satisfied, the hitching assist control unit **40** proceeds to step S222.

(272) In other words, the rear camera **60** captures an unsuitable image of the hitch coupler **30B** to the pattern matching when the vehicle behavior is dynamic. Therefore, when the image is recorded with the vehicle dynamically behaving, the hitching assist control unit **40** determines that the learning has failed and proceeds to step S222, thereby being prevented from erroneously detecting the position of the hitch coupler **30B** due to the pattern matching based on the unsuitable image and reducing the success rate of the hitching assist.

(273) Regarding the content of processing in steps S220 to S223, the hitching assist control unit **40** performs similar processing to the above-described steps, steps S120 to step S123 in the flowcharts illustrated in FIGS. 4 to 8, and therefore the detailed description of the content of the processing in steps S220 to S223 will be omitted here.

(274) In step S224, the hitching assist control unit **40** performs the processing for extracting the trailer **20** from the image of the rear camera **60** to generate the target trajectory, similarly to step S124.

(275) More specifically, the hitching assist control unit **40** determines the trailer region by detecting a combination of a pair of vertical edges emerging at the left and right end portions of the trailer **20** and horizontal edges emerging therebetween.

(276) At this time, when being able to detect one pair of vertical edges and horizontal edges therebetween, the hitching assist control unit **40** carries out pattern matching that extracts a portion most highly correlated to the image pattern of the hitch coupler **30B** from the image region between the one pair of vertical edges as the image region of the hitch coupler **30B** with use of the learned image pattern of the hitch coupler **30B**.

(277) In the processing for extracting the trailer **20** in the above-described step, step S224, the hitching assist control unit **40** limits the region of the extraction processing to around the center of the screen (for example, a range of ± 20 degrees from the center of the screen).

(278) Since the hitching assist control unit **40** presents the display “To request the hitching assist, press the “Hitching Assist” button after confirming that you can view the trailer in the rearview mirror.” toward the driver in the above-described step, step S208, the trailer **20** is highly likely located at a position that allows the driver to visually confirm the trailer **20** in the rearview mirror **10A**, i.e., a position displayed at the center of the screen of the rear camera **60**.

(279) The region around the center of the screen from which the hitching assist control unit **40** extracts the image region of the trailer **20** at this time is a region in which the trailer **20** is displayed in the image of the rear camera **60** when the trailer **20** is located at the position that allows the driver to visually confirm the trailer **20** in the rearview mirror **10A** of the vehicle **10**, i.e., a region in which the trailer **20** is projected on the rearview mirror **10A**.

(280) Then, the hitching assist control unit **40** performs the processing for extracting the trailer **20** limitedly on around the center of the screen, thereby being able to extract the image region of the trailer **20** while reducing the calculation load compared to when extracting the image region of the trailer **20** from the entire image.

(281) Further, when the trailer **20** is not displayed around the center of the screen of the rear camera **60**, this indicates that the hitching assist is highly difficult because the trailer **20** is considerably laterally positionally misaligned from the region behind the vehicle **10**, i.e., the success rate of the hitching assist reduces as the vehicle **10** should be moved back while being significantly steered.

(282) This means that, when the image region of the trailer **20** can be extracted from around the

center of the screen of the rear camera **60**, the vehicle is less required to be moved back while being considerably steered and the hitching assist can be executed with a sufficiently high success rate.

(283) Therefore, in the next step, step **S225**, the hitching assist control unit **40** determines whether the image region of the trailer **20** can be extracted from around the center of the screen. Then, if the image region of the trailer **20** can be extracted (i.e., the present situation satisfies the condition that the trailer **20** can be visually confirmed in the rearview mirror **10A** of the vehicle **10**), the hitching assist control unit **40** proceeds to step **S226** and generates the target trajectory.

(284) In sum, the hitching assist control unit **40** starts the assist in the hitching between the vehicle **10** and the trailer **20** when the trailer **20** is displayed in the predetermined region of the image of the rear camera **60** in the initial state of the hitching and the hitching assist is less difficult without being required to provide a large steering angle in the hitching assist.

(285) On the other hand, if the image region of the trailer **20** cannot be extracted from around the center of the screen, i.e., the success rate of the hitching assist can be estimated to be low with the position of the trailer **20** considerably laterally misaligned, the hitching assist control unit **40** proceeds to step **S232** while skipping steps **S226** and **S227**.

(286) This can prevent the hitching assist control unit **40** from executing the hitching assist with the vehicle **10** and the trailer **20** located in a bad-conditioned positional relationship, thereby preventing the hitching assist from ending up in failure.

(287) Regarding the content of processing in the subsequent steps, steps **S226** to **S232**, the hitching assist control unit **40** performs similar processing to the above-described steps, steps **S125** to step **S131** in the flowcharts illustrated in FIGS. **4** to **8**, and therefore the detailed description of the content of the processing in steps **S226** to **S232** will be omitted here.

(288) After displaying the content “To request the hitching assist, press the “Hitching Assist” button after confirming that you can view the trailer in the rearview mirror.” in step **S208**, the hitching assist control unit **40** can perform the processing for extracting the trailer **20** from the entire screen in step **S224**.

(289) Each technical idea described in the above-described embodiments can be used in combination as appropriate within a range not creating a contradiction.

(290) Having described the contents of the present invention specifically with reference to the preferred embodiments thereof, it is apparent to those skilled in the art that the present invention can be embodied by various modifications based on the basic technical idea and teaching of the present invention.

(291) For example, the hitch that hitches the vehicle **10** and the trailer **20** is not limited to the combination of the hitch ball **30A** and the hitch coupler **30B**.

(292) Further, the condition for learning the image pattern of the hitch coupler **30B** and/or the condition for starting the hitching assist can include a condition that the weather is not a rainfall or snowfall condition, a condition that the illuminance of the region displayed by the rear camera **60** is equal to or higher than a set level, and the like.

(293) Further, it is apparent that the display content on the information display **50** can be changed as appropriate within a range that still allows the driver to understand the intention of the notification.

(294) Further, the notification to the driver is not limited to the display onto the information display **50**, and the hitching assist control unit **40** can issue the notification to the driver by, for example, audio navigation.

(295) Further, the hitching assist control unit **40** can acquire the image pattern of the hitch coupler **30B** by reading out it from a database based on identification information of the trailer **20** or the hitch coupler **30B**.

(296) Further, the hitching assist control unit **40** can re-learn the image pattern of the hitch coupler **30B** based on the image captured by the rear camera **60** while executing the hitching assist.

(297) Further, the hitching assist control unit **40** can also learn an image pattern of the trailer **20**

when learning the image pattern of the hitch coupler **30B**, and can extract the trailer **20** from the image of the rear camera **60** based on the pre-learned image pattern of the trailer **20** when executing the hitching assist.

(298) Further, the hitching assist control unit **40** can move back the vehicle **10** at an appropriate speed in the hitching assist state by outputting an instruction for controlling the driving torque (the vehicle speed) of the vehicle **10** together with the steering instruction in the hitching assist.

(299) Further, the hitching assist control unit **40** can detect that the vehicle **10** and the trailer **20** are hitched manually by the driver based on an operation performed by the driver or an output of a connection sensor that detects whether the hitch ball **30A** and the hitch coupler **30B** are connected, and learn the image pattern of the hitch coupler **30B** based on the image of the rear camera **60** at this time.

(300) Further, in a case where the vehicle **10** includes a four-wheel steering system, the hitching assist control unit **40** can output a steering instruction directed to the front wheels and/or a steering instruction directed to the rear wheels to make the trajectory of the backward movement of the vehicle **10** closer to the target trajectory.

(301) Further, when it is difficult to cause the vehicle **10** to follow the target trajectory by controlling the steering, for example, when the maximum curvature of the target trajectory is larger than a setting value, the hitching assist control unit **40** can cancel the hitching assist and issue a notification prompting the driver to correct the positional relationship between the vehicle **10** and the trailer **20** toward the driver.

(302) Then, technical ideas recognizable from the above-described embodiments will be described in the following description.

(303) A vehicle hitching assist apparatus, as one configuration thereof, is configured to assist in hitching between a trailer including a hitch coupler and a vehicle including a hitch ball connected to the hitch coupler and configured to tow the trailer. From an image acquired by a rear camera configured to capture an image behind the vehicle, the vehicle hitching assist apparatus extracts an image region of the hitch coupler based on a position of the hitch ball from the image when the hitch ball is hidden. At the time of the hitching after that, the vehicle hitching assist apparatus identifies a position of the hitch coupler based on a comparison between the image region and the image acquired by the rear camera, and controls a steering angle of the vehicle so as to move the hitch ball toward the hitch coupler.

(304) In a preferable configuration of the vehicle hitching assist apparatus, the vehicle hitching assist apparatus records the image acquired by the rear camera in the course of the hitching, and repeats processing that searches for the hitch coupler to newly extract the image region from the image that the vehicle hitching assist apparatus temporally dates back to based on the already extracted image region, starting from the image when the hitch ball is hidden, thereby acquiring image regions of the hitch coupler under a plurality of conditions corresponding to different distances from the rear camera to the hitch coupler.

(305) Further, in a preferable configuration of the vehicle hitching assist apparatus, the vehicle hitching assist apparatus extracts the image region of the hitch coupler from the image acquired by the rear camera when a change in a behavior of the vehicle is smaller than a predetermined level.

(306) The present invention shall not be limited to the above-described embodiments, and includes various modifications. For example, the above-described embodiments have been described in detail to facilitate a better understanding of the present invention, and the present invention shall not necessarily be limited to the configuration including all of the described features. Further, a part of the configuration of one embodiment can be replaced with the configuration of another embodiment. Further, one embodiment can also be implemented with a configuration of another embodiment added to the configuration of this embodiment. Further, each of the embodiments can also be implemented with another configuration added, deleted, or replaced with respect to a part of the configuration of this embodiment.

(307) The present application claims priority under the Paris Convention to Japanese Patent Application No. 2018-156122 filed on Aug. 23, 2018. The entire disclosure of Japanese Patent Application No. 2018-156122 filed on Aug. 23, 2018 including the specification, the claims, the drawings, and the abstract is incorporated herein by reference in its entirety.

REFERENCE SIGNS LIST

(308) **10** vehicle **10A** rearview mirror **11, 11** front wheel **12, 12** rear wheel **13** electric power steering apparatus **14** steering control unit (steering controller) **20** trailer **30A** hitch ball (first hitch portion) **30B** hitch coupler (second hitch portion) **40** hitching assist control unit (vehicle hitching assist apparatus or controller) **60** rear camera (external world perception sensor)

Claims

1. An apparatus comprising: a vehicle hitching assist apparatus, the vehicle hitching assist apparatus being arranged in a vehicle equipped with a hitch, the vehicle hitching assist apparatus also being configured to assist in hitching between the vehicle including a first hitch portion and a trailer including a second hitch portion connected to the first hitch portion, wherein the vehicle hitching assist apparatus comprises a microcomputer configured to execute a first mode and a second mode, wherein the first mode configured to, when an image pattern of the second hitch portion is acquired, detect a position of the second hitch portion based on a comparison between the image pattern and a first image acquired by an external world perception sensor mounted on a rear portion of the vehicle, and output a steering instruction for assisting in the hitching between the trailer and the vehicle based on the detected position of the second hitch portion to a steering controller of the vehicle; and the second mode configured to, when the image pattern is not acquired, issue a notification for prompting a driver of the vehicle to manually hitch the trailer and the vehicle, wherein the vehicle hitching assist apparatus learns the image pattern of only the second hitch portion that is provided on the trailer, when the first hitch portion is hidden from the image acquired by the external world perception sensor when the trailer and the vehicle are hitched manually in the second mode, and wherein the vehicle hitching assist apparatus repeats processing that dates back from an immediately preceding image toward the first image among images of the external world perception sensor that have been recorded in a course of the hitching by a driver's operation and extracts a region shaped similarly to an image region of the second hitch portion extracted in the immediately preceding image from the first image as the image region of the second hitch portion, thereby learning the image pattern of the second hitch portion for each of a plurality of conditions corresponding to different distances from the vehicle.
2. The apparatus according to claim 1, wherein the vehicle hitching assist apparatus performs the processing for learning the image pattern based on the first image acquired by the external world perception sensor and a position of the first hitch portion that are recorded when the first hitch portion and the second hitch portion are connected manually in the second mode.
3. The apparatus according to claim 1, wherein the vehicle hitching assist apparatus performs the processing for learning the image pattern based on the first image acquired by the external world perception sensor when a change in a behavior of the vehicle is smaller than a predetermined level.
4. The apparatus according to claim 1, wherein the vehicle hitching assist apparatus issues the notification for prompting the driver of the vehicle to change a condition of the manual hitching, when the processing for learning the image pattern has failed.
5. The apparatus according to claim 1, wherein the vehicle hitching assist apparatus records the first image acquired by the external world perception sensor when the first hitch portion and the second hitch portion are connected manually in the second mode, and starts the processing for learning the image pattern upon completion of a manual connection between the first hitch portion and the second hitch portion.
6. The apparatus according to claim 1, wherein the vehicle hitching assist apparatus starts assisting

in the hitching between the trailer and the vehicle with the trailer displayed in a predetermined region of the first image acquired by the external world perception sensor in the first mode.

7. The apparatus according to claim 6, wherein the predetermined region is a region in which the trailer is projected on a rearview mirror of the vehicle.

8. A method comprising: providing a vehicle; providing a trailer that is connectable to the vehicle; mounting an external world perception sensor on a rear portion of the vehicle; arranging a hitching assist apparatus in the vehicle, wherein the hitching assist apparatus is configured to assist in hitching between a vehicle including a first hitch portion and the trailer including a second hitch portion connected to the first hitch portion, detecting a position of the second hitch portion based on a comparison between an image pattern of the second hitch portion and a first image acquired by the external world perception sensor and outputting a steering instruction for assisting in the hitching between the trailer and the vehicle based on the detected position of the second hitch portion to a steering controller of the vehicle, when the image pattern of the second hitch portion is acquired; and issuing a notification for prompting a driver of the vehicle to manually hitch the trailer and the vehicle, when the image pattern is not acquired, wherein the vehicle hitching assist method further comprising: learning the image pattern of only the second hitch portion provided on the trailer, when the first hitch portion is hidden from the first image acquired by the external world perception sensor when the trailer and the vehicle are hitched manually in the second mode, wherein the learning includes: repeating processing that dates back from an immediately preceding image toward the first image among images of the external world perception sensor that have been recorded in a course of the hitching by a driver's operation, and extracting a region shaped similarly to an image region of the second hitch portion extracted in the immediately preceding image from the first image as the image region of the second hitch portion, thereby learning the image pattern of the second hitch portion for each of a plurality of conditions corresponding to different distances from the vehicle.

9. A system comprising: a vehicle; a controller that is mounted on the vehicle; a trailer that is connectable to the vehicle; and an external world reception sensor mounted on a rear portion of the vehicle, wherein the controller is configured to: i) assist in hitching between the vehicle including a first hitch portion and a trailer including a second hitch portion connected to the first hitch portion, ii) when an image pattern of the second hitch portion is acquired, detect a position of the second hitch portion based on a comparison between the image pattern and a first image acquired by the external world perception sensor, iii) output a steering instruction for assisting in the hitching between the trailer and the vehicle based on the detected position of the second hitch portion to a steering controller of the vehicle, and iv) when the image pattern is not acquired, issue a notification for prompting a driver of the vehicle to manually hitch the trailer and the vehicle, and wherein the vehicle hitching assist apparatus learns the image pattern of only the second hitch portion provided on the trailer when the first hitch portion is hidden from the first image acquired by the external world perception sensor when the trailer and the vehicle are hitched manually in the second mode wherein the vehicle hitching assist apparatus repeats processing that dates back from an immediately preceding image toward the first image among images of the external world perception sensor that have been recorded in a course of the hitching by a driver's operation and extracts a region shaped similarly to an image region of the second hitch portion extracted in the immediately preceding image from the first image as the image region of the second hitch portion, thereby learning the image pattern of the second hitch portion for each of a plurality of conditions corresponding to different distances from the vehicle.
